# Digital Twin of an Electrical Motor Based on Empirical Performance Model

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***Abstract*— The concept of Digital Twin is creating and maintaining a digital representation of the real physical entity and supporting its performance utilizing simulation and optimization tools, which are fed with real data. Development and implementation of Digital Twin technology is a hot topic in many industry-oriented research projects. However, more detailed studies are needed on design methods for Digital Twins, which allows full synchronization and connectivity between virtual and real environments. This paper presents a development case study of Digital Twin for an electric motor based on an empirical performance model. The paper discusses the way how the data required for the Digital Twin were obtained and present a detailed structural description of the virtual entity. Unity3D is used for physics simulations and visualization of the Digital Twin, because of simple accessible and meticulously detailed documentation highly useful for rapid prototyping in early stages.**

***Keywords*— *Motor drives, Model-driven development, Industry applications***

## I. Introduction

The concept of Digital Twin (DT) is creating and maintaining a digital representation of the real physical entity and supporting its performance through simulation and optimization tools, which are fed with real and updated data. DT is a part of the digitalization and simulation pillar of Industry 4.0 paradigm [1]. The application of Industry 4.0 technologies in many sectors has been barely examined yet since the variation of sub-processes and manufacturing technologies results in a large number of potential combinations. DT technology may present a valuable tool for the rapid integration of Industry 4.0 in today's technologies. DT covers three components: knowledge content, effect, and functionality, and application domain [2]; for utilization of a DT technology, the relations of the three components must be recognized, and the gaps remaining for exploration identified.

DT consist of three main components – the physical entity in the real world, its virtual entity (or virtual model(s)), and the data, furthermore services and connections are presented as separate entities. Generally, a real physical entity consists of assorted subsystems and sensory devices; however, for the establishment of a virtual entity, it is possible to use already collected data. The virtual entity subsists of one or several models according to the DT application; it can be the spatial, behavior, thermal, rule, or another model. The DT data contain various sets of data from the real physical entity that can be obtained by the implementation of different sensors and data acquisition platforms. The service entity mainly includes regulation for both virtual and physical entities and may carry several sub-services, such as maintenance and diagnostic, energy optimization, path planning, etc. The connection entity usually defines the interaction between other entities. More studies are needed on design methods for DT, which allows full synchronization and connectivity between virtual and real environments [3].

A. Rasheed et al. [4] give a taste of what a state-of-the-art DT of an offshore oil platform looks like. The DT at the example study is continuously updated with sensors data in near real-time and can be supported by synthetic data generated from a virtual entity. In this case, DT does not only give real physical entity real-time information for more precise control and decision making but may serve as prediction makers about how the asset will evolve or behave in the future. A perfect DT will be identical to the real physical entity in terms of appearance and behavior with the extra advantage of making future predictions. The DT presented in the study also provide the possibility for operator or scholar to interact with the physical entity using an avatar.

To create an industrial avatar in Mixed Reality (XR) application with realistic physics as a part of a DT, various software packages can be used. For it, game engines such as Unity3D, UnrealEngine, and CryEngine are popular tools used for creating industrial simulations with real-time graphics requirements. For simulations that are more precise or mathematical or realistic task, other toolkits can be used as an addition on separately, as MatLab, Visual Components Arena, Technomatrix.

There are some particular reasons for using game engines as visualization output tools:

1. The easiest way to implement graphical applications and make an animation from external commands or internal algorithms.
2. Out-of-the-box support of existing XR hardware and embedded connectivity methods for various protocols.
3. Community support and large knowledge bases both from consumers, experts in the fields, and various field researchers.
4. Flexible to work with real physical external equipment by telemetry connection means.

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Industrial and research applications created using those engines are called "serious games."[5] Each of those visual engines has its advantages and drawbacks – in the end, the selection is made according to specific task requirements and user competence level.

Except for game engines, there exist other software products used for industrial simulation applications:

- MATLAB – accessible simulation environment and programming language that can be used for running physical simulations connected to game engine code;
- Computer-aided design (CAD) and 3D modeling software Solidworks, ANSYS, COMSOL Multiphysics, etc. that typically used for physical design and optimization;
- Visual Components, Technomatrix – the software's for creating interactive factory layouts and plans; Processes visualizations;
- Industrial Robot offline programming software packages – latest versions of robot programming environments by ABB, Yaskawa, Kuka, and other companies have simulation support;
- Open-source software packages as a Robot Operating System (ROS) embedded with simulation and visualization tools as Gazebo, Rviz.

In the current study, Unity3D is used for physics simulations and visualization of the DT, because of previous research stuff experience and high-quality documentation provided by software producer and community.

A recent study presents a part of the project that aimed to develop a specialized unsupervised prognosis and control platform for electric propulsion drive systems performance estimation of an autonomous self-driving electric vehicle. This goal requires the development of several subtasks, and the related objectives, one of them is to develop physical models of different energy system components (motors, gearboxes, power converters, etc.) and the related reduced models of these components (testbed), which will serve to construct the DT of the system [6].

The paper is structured as follows: Section II reports how the empirical performance model was obtained. Section III presents the detailed structure description of the Digital Twin virtual entity ant its implementation framework. IN the discussion and conclusion section, the further development and application of presented DT are characterized.

## II. Empirical Performance Model

In order to have a reasonable empirical performance model, an induction motor (IM) ABB 3GAA132214-ADE was studied. The tested electrical motor is driven by an industrial frequency converter (ABB ACS880) with Direct Torque Control (DTC) algorithm. Additional frequency converter (ABB ACS800) IM setup was used as a load. To obtain an efficiency map, the load torque is gradually increased from zero to the calculated during the design stage rated load torque in a present ramp mode. All the experiments are performed in a real-time setup, including motors and frequency converters. The process of efficiency map obtaining is described in previous research by authors [7].

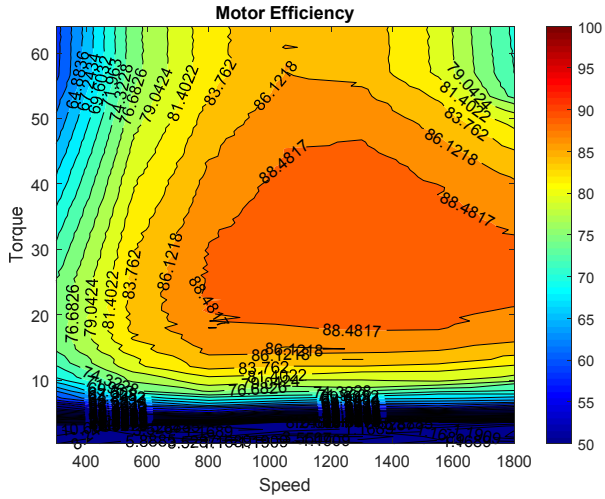


Fig. 1. The empirical performance model of induction motor used for Digital Twin.

The ratio of the motor shaft power ($P_{mech_motor}$) to the total electrical input power ($P_{in_total}$) is taken as motor efficiency:

$$\eta_{\text{motor}} = P_{mech_motor}/P_{in_total} \quad (1)$$

The graphical representation of an empirical performance model of IM is presented in Fig. 1 and rated data of the motor is given in Table I.

TABLE I. Rated data of induction motor

| Parameter | Unit | Value |
|---|---|---|
| Motor frame size | | 132 MA |
| Rated Power | kW | 7.5 |
| Rated Current | A | 22 |
| Rated Speed | rpm | 1460 |
| cos• | | 0.6 |
| Moment of inertia | kgm² | 0.048 |

The empirical performance model characterizes the loss distribution of IM on speed-torque characteristics. The copper losses are the dominant power losses in any electrical motor [7]. The torque of IM is proportional to the electrical loading and the magnetic loading. To increase the torque produced by the motor, both (electrical and magnetic) loadings must be increased. However, due to the saturation of the stator iron core in high magnetic flux density, the only constructive way to increase the motor torque is by increasing the electrical loading. That can be achieved by increasing the current density in motor windings, which in turn higher copper losses as well as lower efficiency.

Numerical representation of an IM empirical performance model may be used as one of the inputs for building DT to evaluate the IM performance over a given speed-torque region.

## III. Digital Twin

### A. *Structure of the Digital Twin*

The DT for the IM is set up in the following way: Unity 3D physics engine is used to simulate fundamental physics while connected via Robot Operation System (ROS) bridge, Linux ROS nodes simulate more advanced electrical machine-specific behaviors such as motor efficiency following the efficiency map and motor controller. As shown in Fig. 2. The primary benefit of Unity 3D, that it has a very

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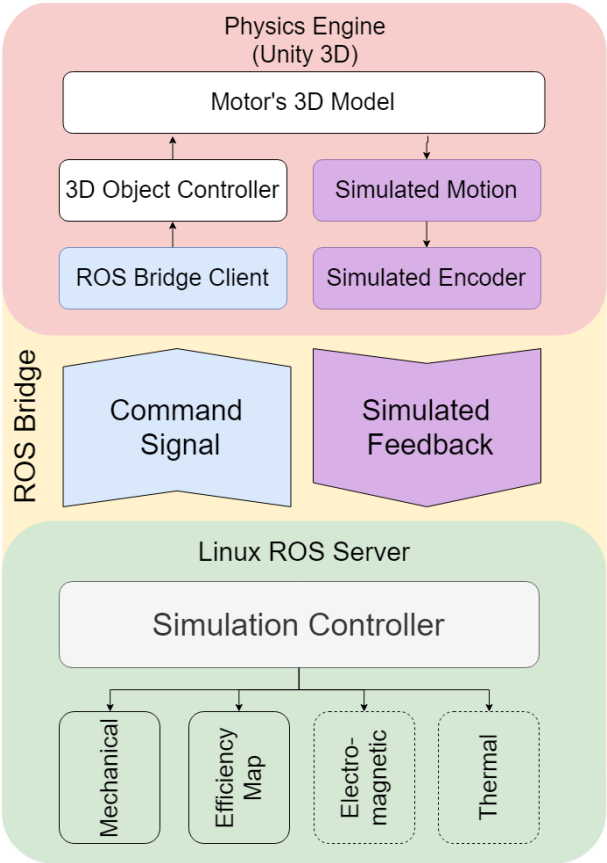


Fig. 2. The operational architecture of Digital Twin.

attractive system to work with 3D objects and simulate the effects of virtual physical forces. These upsides were a primaryreason for choosing Unity3D over more commonly used in research MATLAB, Matplotlib, or Mathworks on the recent stage of the project.

Linux ROS Server sends a User Datagram Protocol (UDP) command packet via ROS Bridge standard to the Unity 3D Visualizer. Its main task is to receive a control message from a specified IP address in a standardized ROS Bridge message format that is being published by the Linux ROS Server and control the 3D models to display the behavior of the motor as it would be in real life. It is also responsible for sending feedback to the Linux ROS Server for processing, just like a real encoder would send updates on the motor rotation rate to a motor controller.

Linux ROS Server Consists of multiple nodes that simulate various aspects (e.g., mechanical, thermal, electrical, diagnostic, etc.) of the motor, a single motor controller simulator node and a Simulation controller node that combines information coming from all other aspect simulation nodes and feedback from the Unity 3D, process it and send the data to a visualization client in Unity 3D.

More in the details physics engine of the DT is represented in Fig. 3. Unity 3D Visualizer message is received in the form of a UDP package in a custom ROS Bridge format by the ROS Bridge Client, reserialized into input variables (Empirically Estimated Velocity and Torque) of the motor and passed on to the 3D Object Controller as variables. Object Controller converts data from the empirical performance model to DT model velocity at which the shaft of the motor rotates. This data is applied to the motor's 3D visualization. Then a simulated encoder (so-called Virtual

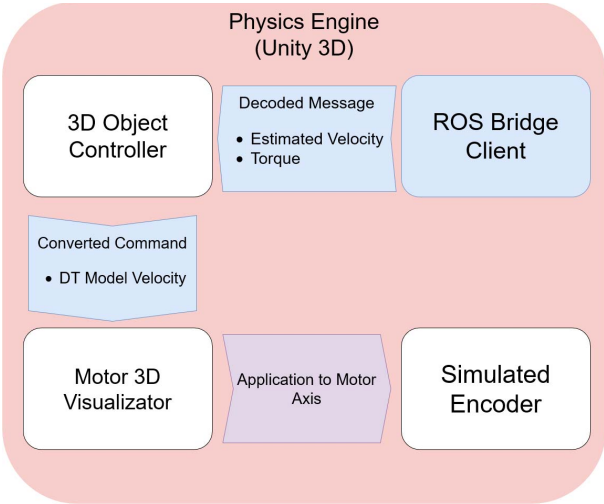


Fig. 4. Detailed representation of Digital Twin Physics Engine.

Sensor) takes the angular velocity of the motor shaft, converts it to feedback data, and sends it to the Linux ROS Server.

Currently, there are 2 active nodes in the simulation: a motor controller simulator node and a mechanical simulation node. The motor controller, has a proportional–integral–derivative (PID) controller that provides a control signal to the motor. Mechanical simulation node works with an efficiency map measured and recorded on a real motor and applies it to a simulation to emulate proper torques and power output at certain angular velocity to make DT act just like real entity motor.

### *B. Graphical representation or spatial model*

The DT model does not undoubtedly mean a spatial or graphical model. The main accent should be paid to process flow and relations behind it as well as the data entity. However, the application of augmented reality (AR) and virtual reality (VR) tools for distance simulation via DT add safety layer and more possibilities for accelerated lifecycle tests, applications in hazardous environments, and remote work maintenance overall. Implementation of a virtual visual model can prevent plenty number of mistakes and errors without any performance or economic loss. Besides that, such a model can be a valuable tool for the education of technical staff and in universities. In the spatial model, the components of a real physical entity (e.g., windings, rotor, shaft, encoder, etc.) are constructed as CAD geometric models to be assembled in the virtual entity. The spatial model used for DT visualization is shown in Fig. 4.

Physics Engine has a ROS Bridge client that receives a data package from ROS Server. The data in that package is

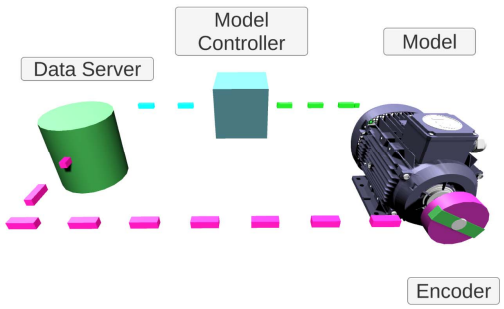


Fig. 3. The spatial model used for Digital Twin visualization.

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extracted by the ROS Bridge client and sent to the 3D object controller that recalculates new positions and orientations of the simulated motor components. It then updates the 3D model with new states. This step ensures that a 3D model is acting like a real motor would from a visual standpoint. Then model takes into account any other physical forces applied to the motor, such as friction losses and moment of inertia of the motor simulated via angular drag feature in Unity 3D's Rigidbody, simulation and updates simulated encoder values which are then packed and sent to the ROS Server to act as feedback for the speed controller. The simulated encoder is modular and can be replaced by any other necessary feedback simulation. Currently a PID controller is used for its simplicity and ease of use.

## IV. Discussion

The term Digital Siblings (DS) is also introduced in [4] and can be considered as copy(ies) of the physical entity which need not necessarily run in real-time but can be used to test out hypothetical scenarios for maintenance, diagnostics, "what if?" analysis and risk assessment. Reduced models of the physical entities for DT and DS can be constructed using different model order reduction methods. A variety of condition monitoring techniques are available nowadays and may be combined with DT and DS approaches, e.g. electromagnetic field monitoring, noise and vibration monitoring, infrared recognition, temperature measurements, radio frequency emission monitoring, chemical analysis, acoustic noise measurement, motor current signature analysis (MCSA) and most advanced artificial intelligence-based techniques, such as fuzzy logic and neural networks etc. [8]

Very different software can be combined to achieve the proper sibling. An example of hybrid Finite Element Analysis (FEA)-Simulink modelling of permanent magnet assisted synchronous reluctance motor is presented by J. Pando-Acedo et al. in [9]. Cross-platform software that combines multiphysics graphical applications with powerful pre-processing, solvers, and post-processing capabilities will be preferable for DS and DT creation. Nowadays, some of the optimization framework as Artap [10] are linking together different software parts and interfaces for both of gradient-free and evolutionary based optimization methods, meta-modeling techniques, surrogate models and FEM solvers.

That fact that developed DT ROS nodes are modular makes it very easy to add nodes responsible for electromagnetic, thermal, diagnostic or any other simulations. Different reduced models of the devices running parallel and in real-time can be used to assemble a DT. Studies shows [11] that model order reduction is a promising tool for controlling industrial processes, where some of the parameters cannot be measured directly.

Diagnostics as a promising application will be considered as foreground implementation of the DT based on the modified winding function-based model described in [12]. However,

## V. Conclusion

The concept of DT is gaining popularity nowdays in many different industry oriented fields. Creating and maintaining a digital representation of the real physical entity, and supporting device performance using different simulation and optimization tools are a key goal of many research works.

This paper presents a development case of DT for an electric motor based on an empirical performance model. The paper consider obtained data required for the DT development. A detailed structural description of the virtual entity based on Unity3D engine is presented.

The main application of the recent DT is a loading motor-drive system for the test bench to estimate the performance of the electric propulsion drive system of an autonomous electric vehicle. The development and implementation of the concept of DT, will help to provide a brand-new approach for measurement and estimation of the performance of motor-drive systems.

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